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Reinsurance Type Inventory — Calculated Using TSRe Fields

Every structure from the standalone reference, with each formula's terms mapped to the actual TSRe field(s) that store them. Where TSRe has **no dedicated field** for a mechanic, that's stated explicitly — several of these structures (drop-down, cascading, top-and-drop, outer aggregate deductibles) aren't a single field at all in TSRe; they're **built by combining multiple treaty/program records through inuring and sourcing relationships**, exactly as the AIR reference document itself describes.

Field key: tRT = tReinsuranceTreaty · tRP = tReinsuranceProgram · tRPSI = tReinsurance-ProgramSourceInure · CBAV = tCatBondAttributeValue (free-text attributes) · ILST = ILSTerms (RiskAnalyzer-side; noted only where TSRe itself has no equivalent).

Part 0: Foundational Mechanics — TSRe Fields

Concept	TSRe field(s)
Occurrence Retention	tRT.OccurrenceRetention
Occurrence Limit	tRT.OccurrenceLimit
Aggregate Retention	tRT.AggregateRetention
Aggregate Limit	tRT.AggregateLimit
Per-risk Retention/Limit	tRT.RiskRetention, tRT.RiskLimit
Reinstatement count	tRT.Reinstatement
Reinstatement premium handling (by amount / by time)	tRT.ReinstProvisionalAmount, tRT.ReinstProvisionalTime
Reinstatement premium (actual)	tRT.ReinstatementPremium (also per-reinstatement detail in tReinsurance-TreatyReinstatement.ReinstatementPremium, .AmountProvision, .TimeProvision)
Coinsurance	tRT.CoinsuranceAmount
Gross / Net / Participation 3 / Participation 4	tRT.ParticipationGross, ParticipationNet, Participation3, Participation4
Percent Placed	tRT.PlacedAmount
Ceded %	tRT.CededAmount (treaty-wide) or tAppliesToAreaGeoXRef.CededPercent (by specific geography)
Inuring relationship	tReinsuranceProgramSourceInure — ReinsuranceProgramSubjectSID, ReinsuranceProgramSourceInureSID, IsSource, IsInuringNextProgramFlow, ParticipationAmount
Inuring order/sequence	tRT.InuringSequenceNumber
Starting event number	tRT.StartingEventNumber

$\text{Eligible Loss (event)} = \min(\max(0, \text{Loss} - \text{tRT.OccurrenceRetention}), \text{tRT.OccurrenceLimit})$
 $\text{Agg Retention Remaining} = \max(0, \text{tRT.AggregateRetention} - \sum \text{prior Eligible Losses})$
 $\text{Contract Loss} = \min(\max(0, \text{Eligible Loss} - \text{Agg Retention Remaining}), \text{tRT.AggregateLimit})$
 $\text{Reinsurer's share} = \text{Layer Loss} \times (1 - \text{tRT.CoinsuranceAmount})$
 $\text{Reinsurer's Loss} = 100\% \text{ Contract Loss} \times \text{tRT.ParticipationGross (or ParticipationNet/3/4)}$
 $\text{Net Retained Loss} = \text{Company Loss} - \text{Inuring Program Loss}$
 $\quad - (100\% \text{ Contract Loss} \times \text{tRT.PlacedAmount})$
 $\text{Aggregate Limit} = \text{tRT.OccurrenceLimit} \times (1 + \text{tRT.Reinstatement})$
 $\text{Reinstatement Premium} = \text{tRT.Premium}$
 $\quad \times (\text{Limit Reinstated} / \text{tRT.OccurrenceLimit})$
 $\quad \quad \text{-- driven by ReinstProvisionalAmount}$
 $\quad \times (\text{Time Remaining} / \text{Full Period})$
 $\quad \quad \text{-- driven by ReinstProvisionalTime}$

Part 1: Proportional Structures

1.1 Quota Share

Term	TSRe field
Cession %	tRT.ParticipationNet/ParticipationGross, or tAppliesToAreaGeoXRef.CededPercent if it varies by geography
Occurrence/Aggregate caps	tRT.OccurrenceLimit, tRT.AggregateLimit

$\text{Contract Loss} = \min(\text{tRT.ParticipationGross} \times \text{Company Loss}, \text{tRT.OccurrenceLimit})$ [per event]
 $\text{Contract Loss (year)} = \min(\sum \text{above}, \text{tRT.AggregateLimit})$

1.2 Surplus Share

No dedicated TSRe field for “retained line” or “number of lines.” TSRe’s schema has no per-risk cession-scaling concept in the tables reviewed — this would have to be approximated using tRT.RiskLimit/RiskRetention (the per-risk fields) if TSRe supports true surplus-share cession elsewhere, but nothing in the schema/SQL reviewed distinguishes it from a per-risk XOL.

$\text{Cession \% (per risk)} = \max(0, (\text{Risk Sum Insured} - [\text{no TSRe field}]) / \text{Risk Sum Insured})$
 $\text{Reinsurer's Loss} = \text{Cession \%} \times \text{Risk-Level Loss}$

Part 2: Excess of Loss Structures

2.1 Per-Risk XOL

Term	TSRe field
Retention / Limit	tRT.RiskRetention, tRT.RiskLimit

$\text{Recovery} = \min(\max(0, \text{Risk Loss} - \text{tRT.RiskRetention}), \text{tRT.RiskLimit})$

2.2 Per-Occurrence (Catastrophe) XOL

Term	TSRe field
Retention / Limit	tRT.OccurrenceRetention, tRT.OccurrenceLimit
Reinstatements	tRT.Reinstatement
Coinsurance	tRT.CoinsuranceAmount

Recovery = min(max(0, Occurrence Loss – tRT.OccurrenceRetention), tRT.OccurrenceLimit)

2.3 Aggregate XOL

Term	TSRe field
Aggregate Retention / Limit	tRT.AggregateRetention, tRT.AggregateLimit

Contract Loss = min(max(0, Σ Loss – tRT.AggregateRetention), tRT.AggregateLimit)

2.4 Excess of Loss with (Outer) Aggregate Deductible

Not a single-field mechanic — built as two linked TSRe programs, exactly as documented:

Component	TSRe representation
“Deductible absorption” program	A separate tReinsuranceProgram/tRT (e.g., a Quota Share with AggregateLimit set to the deductible amount)
Relationship	tReinsuranceProgramSourceInure — the deductible program IsSource, and inures to the benefit of the XOL program
The XOL layer itself	tRT.OccurrenceRetention, tRT.OccurrenceLimit on the second program

Absorbed = min(Σ Ground-Up Losses, [Deductible Program].tRT.AggregateLimit)

Remaining Loss = Ground-Up Loss – Absorbed
(via tReinsuranceProgramSourceInure relationship)

Contract Loss = min(max(0, Remaining Loss – tRT.OccurrenceRetention), tRT.OccurrenceLimit)

2.5 Excess of Loss with Otherwise Recoverable (Inner Aggregate) Deductible

This one **is** a direct single-treaty field mechanic — it’s just the standard Aggregate Retention applied in its normal sequence (Part 0), on the *same* treaty record as the occurrence terms:

Term	TSRe field
Occurrence Retention / Limit	tRT.OccurrenceRetention, tRT.OccurrenceLimit
Aggregate Retention (the “inner” deductible) / Limit	tRT.AggregateRetention, tRT.AggregateLimit

Eligible Loss = min(max(0, Loss – tRT.OccurrenceRetention), tRT.OccurrenceLimit)

Agg Retention Remaining = max(0, tRT.AggregateRetention – Σ prior Eligible Losses)

Contract Loss = min(max(0, Eligible Loss – Agg Retention Remaining), tRT.AggregateLimit)

2.6 Franchise Deductible

No dedicated “franchise” field in TSRe — built using the industry/company loss trigger fields as the threshold gate, combined with the occurrence limit:

Term	TSRe field
Trigger threshold	tRT.CompanyLossMinimum (or IndustryLossMinimum for an index-based version)
Payout limit	tRT.OccurrenceLimit
$\text{Contract Loss} = 0 \quad \text{if Loss} < \text{tRT.CompanyLossMinimum}$ $\text{Contract Loss} = \min(\text{Loss}, \text{tRT.OccurrenceLimit}) \quad \text{if Loss} \geq \text{tRT.CompanyLossMinimum}$	

2.7 Reverse Franchise Deductible

Same fields as 2.6, but using the *maximum* field as a ceiling instead of the minimum as a floor:

Term	TSRe field
Trigger ceiling	tRT.CompanyLossMaximum (or IndustryLossMaximum)
Payout limit	tRT.OccurrenceLimit
$\text{Contract Loss} = \min(\text{Loss}, \text{tRT.OccurrenceLimit}) \quad \text{if Loss} \leq \text{tRT.CompanyLossMaximum}$ $\text{Contract Loss} = 0 \quad \text{if Loss} > \text{tRT.CompanyLossMaximum}$	

2.8 Drop-Down Cover

Not one field — a second treaty record (a “step-layer”) sharing the same program, activated once the first is exhausted:

Term	TSRe field
Layer 1 terms	tRT.OccurrenceRetention, tRT.OccurrenceLimit, tRT.Reinstatement (on the primary treaty row)
Step-layer terms	Same fields, on a second tRT row; StepLayerSequenceNumber links it as the drop-down step

Layer 1: Recovery = $\min(\max(0, \text{Loss} - \text{tRT}_1.\text{OccurrenceRetention}), \text{tRT}_1.\text{OccurrenceLimit})$
 Step-Layer: Recovery = $\min(\max(0, \text{Loss} - \text{tRT}_2.\text{OccurrenceRetention}), \text{tRT}_2.\text{OccurrenceLimit})$
 [tRT₂.OccurrenceRetention < tRT₁.OccurrenceRetention;
 activates once tRT₁ is exhausted]

2.9 Nth-Event Cover

Method A (aggregate-retention sizing) — pure field mechanic:

Set tRT.AggregateRetention = tRT.OccurrenceLimit × (N – 1)
 [then the standard Part 0 four-step formula applies]

Method B (hard sequence filter) — a direct, dedicated field:

$\text{Contract Loss} = 0$ for Event Sequence < tRT.StartingEventNumber
 $\text{Contract Loss} = \min(\max(0, \text{Loss} - \text{tRT.OccurrenceRetention}), \text{tRT.OccurrenceLimit})$
 for Event Sequence $\geq$ tRT.StartingEventNumber

tRT.StartingEventNumber is confirmed **mapped through to ILSTerms.StartingEventNumber** on the RiskAnalyzer side — but (per the earlier gap analysis in this conversation) never actually *referenced* by the risk-statistics calculation reviewed, so this field currently has no effect on computed results even though it's correctly captured.

2.10 Cascading Structure

Built entirely from relationships, not a formula field — each layer is its own tRT row, linked via tReinsuranceProgramSourceInure/InuringSequenceNumber:

Term	TSRe field
Each layer's own terms	tRT.OccurrenceRetention, tRT.OccurrenceLimit (one row per layer)
Cascade order	tRT.InuringSequenceNumber
The inuring relationship itself	tReinsuranceProgramSourceInure

$\text{Layer 1 Loss} = \min(\max(0, \text{Event Loss} - \text{tRT}_1.\text{OccurrenceRetention}), \text{tRT}_1.\text{OccurrenceLimit})$
 $\text{Layer 2 Loss} = \min(\max(0, (\text{Event Loss} - \text{Layer 1 Loss}) - \text{tRT}_2.\text{OccurrenceRetention}), \text{tRT}_2.\text{OccurrenceLimit})$
 $\text{Layer 3 Loss} = \min(\max(0, (\text{Event Loss} - \text{Layer 1 Loss} - \text{Layer 2 Loss}) - \text{tRT}_3.\text{OccurrenceRetention}), \text{tRT}_3.\text{OccurrenceLimit})$

2.11 Top-and-Drop Structure

Also a relationship-built structure — same fields as Cascading (2.10) or Drop-Down (2.8), combined:

IF First Event exhausts the lower layers (per tReinsuranceProgramSourceInure chain):
 Top Layer Loss = $\min(\max(0, \text{Remaining Loss} - \text{tRT}_{\text{top}}.\text{OccurrenceRetention}), \text{tRT}_{\text{top}}.\text{OccurrenceLimit})$
 ELSE (top layer untouched, drops to act as a lower layer for the next event):
 Top Layer Loss (2nd event) = $\min(\max(0, \text{2nd Event Loss} - \text{tRT}_{\text{top}}.\text{OccurrenceRetention}), \text{tRT}_{\text{top}}.\text{OccurrenceLimit})$

Part 3: Index & Parametric Structures

3.1 Industry Loss Warranty (ILW)

Term	TSRe field
Industry-index trigger	tRT.IndustryLossMinimum, tRT.IndustryLossMaximum
Company-footprint trigger (proxy exposure)	tRT.CompanyLossMinimum, tRT.CompanyLossMaximum
Payout terms	tRT.OccurrenceRetention, tRT.OccurrenceLimit

$\text{Trigger Met} = (\text{Industry Loss} \geq \text{tRT.IndustryLossMinimum})$
 [or $\text{Company Loss} \geq \text{tRT.CompanyLossMinimum}$]
 $\text{Contract Loss} = \min(\max(0, \text{Loss} - \text{tRT.OccurrenceRetention}), \text{tRT.OccurrenceLimit})$ IF Trigger Met, ELSE 0

Confirmed important gap: IndustryLossMinimum/Maximum and CompanyLossMinimum/Maximum are confirmed **mapped** into ILSTerms on the RiskAnalyzer side, but confirmed **never referenced** by the risk-statistics calculation reviewed — so even when these trigger fields are correctly populated from TSRe, the actual index-comparison logic isn't applied downstream; the calculation just uses the treaty's own simulated net loss.

3.2 Parameter-Based ILW

No TSRe field found for physical event parameters (magnitude, landfall pressure, epicenter region) anywhere in the schema reviewed — TSRe's tAppliesToEventFilterRule table (Parameter, Condition, Value1, Value2) is structurally exactly what this needs (a rule-based event filter), and is the correct TSRe location for this mechanic, but it was not referenced by name in any conversion script reviewed, so its actual population/usage in the pipeline is unconfirmed.

Trigger Met = (tAppliesToEventFilterRule.Value1 condition met, e.g. Magnitude ≥ 7.0)
 Contract Loss = $\min(\max(0, \text{Loss} - \text{tRT.OccurrenceRetention}), \text{tRT.OccurrenceLimit})$ IF Trigger Met, ELSE 0

Part 4: Capital Markets / ILS Structures

4.1 CAT Bond

Term	TSRe field
Principal / notional	tRT.PrincipalAmount
Annual spread	tRT.PercentAnnualSpread
Reference rate	tCatBondAttributeValue where attribute = "Reference Rate"
Trigger type	tRT.TriggerTypeCode (coded) — though confirmed the actual RiskAnalyzer-side ILSTerms.TriggerTypeCode is sourced from a CAT bond attribute (hardcoded attribute ID 62) instead of this field

Coupon = (Reference Rate + tRT.PercentAnnualSpread) $\times$ tRT.PrincipalAmount
 Principal Reduction = $\min(\text{Loss}, \text{tRT.PrincipalAmount})$
 Full Exhaustion = Loss $\geq$ tRT.PrincipalAmount

4.2 Sidecar

Term	TSRe field
Participation %	tRT.ParticipationNet/ParticipationGross
Aggregate Limit (= collateral)	tRT.AggregateLimit

Investor's Loss = tRT.ParticipationGross $\times$ Segment's Net Loss,
 capped at tRT.AggregateLimit

4.3 Collateralized Reinsurance (CRI)

No separate fields — uses whichever formula (1.1, 2.2, or 2.3) matches its actual attachment basis. TSRe's tRT.ReinsuranceTypeCode is the field that would indicate this (confirmed values seen include AggXOL, QS, RPP, and CATB).

4.4 RPP

tRT.ReinsuranceTypeCode = 'RPP' is a confirmed *value* that exists and is checked in downstream logic, but neither its full name nor its distinct calculation mechanic (as opposed to a plain proportional or aggregate structure) was found anywhere in the TSRe schema documentation or the AIR reference document reviewed. No formula can be honestly given beyond: it's treated identically to AggXOL/QS wherever it appears in the code reviewed.

Summary: Field Coverage Across All 14+ Structures

Structure	Has dedicated TSRe field(s)?	Built from relationships instead?	Field exists but unused downstream?
Quota Share	Yes	—	Participation not read back out (uses AAL heuristic instead)
Surplus Share	No	—	—
Per-Risk XOL	Yes	—	Not referenced in risk-statistics calc at all
CAT XOL	Yes	—	Retention not reapplied downstream; reinstatements not modeled
Aggregate XOL	Yes	—	—
Outer Aggregate Deductible	—	Yes (2 linked programs)	—
Inner Aggregate (“Otherwise Recoverable”) Deductible	Yes	—	—
Franchise / Reverse Franchise Deductible	Partial (via Loss Min/Max)	—	—
Drop-Down Cover	—	Yes (step-layer)	—
Nth-Event Cover	Yes (StartingEventNumber)	Also via Agg Retention sizing	Confirmed mapped but never referenced downstream
Cascading Structure	—	Yes (inuring chain)	—
Top-and-Drop	—	Yes (inuring + step-layer)	—
ILW	Yes (Industry/Company Loss Min/Max)	—	Confirmed mapped but never referenced downstream
Parameter-Based ILW	Possibly (tAppliesToEventFilterRule)	—	Usage unconfirmed
CAT Bond	Yes	—	Blocked by the 'CAT Bond'/'CATBond' string bug (see earlier analysis)
Sidecar	Yes	—	Blocked by the same bug
Collateralized Re (CRI)	Via ReinsuranceTypeCode routing	—	Routing itself is unreachable (same bug)

Structure	Has dedicated TSRe field(s)?	Built from relationships instead?	Field exists but unused downstream?
RPP	Value exists, mechanic undocumented	—	—

The pattern holds across every structure type, not just the ones examined earlier in this conversation: TSRe’s schema is rich enough to represent nearly the entire market taxonomy — either through dedicated fields (proportional and excess-of-loss basics) or through its relationship tables (tReinsuranceProgramSourceInure, InuringSequenceNumber) for the more exotic structures (cascading, top-and-drop, outer deductibles). The recurring gap isn’t in what TSRe can express — it’s that several fields which **are** correctly captured and mapped through to RiskAnalyzer’s ILSTerms are then never read by the actual risk-statistics calculation, and the one field that would activate type-specific logic (Security_or_Reinsurance_Type) is hardcoded wrong for every instrument the reviewed pipeline produces.